DATA MANAGEMENT CARPENTRY · EPISODE 4

Documenting Your Data

Recording provenance, designing effective README files, and embedding metadata so research remains understandable and reproducible.

EPISODE SCOPE

Questions & Objectives

Core Questions

How do I remember where data originated?

What information should I document?

Where should documentation be stored?

What makes an effective README file?

Learning Objectives

Explain the purpose of data documentation.

Identify key metadata to capture.

Create structured project README files.

Document data provenance and workflows.

CONTEXT

Navigation vs. Understanding

What We Have Solved

Our folder structures are tidy and our filenames follow consistent, sortable conventions.

> **Good structures help people NAVIGATE data.**

Unanswered Questions Remain

- Who collected this, and when?
- What do the variables and units mean?
- Which files are original raw inputs vs. outputs?
- What software tools and versions were used?

> **Documentation helps people UNDERSTAND data.**

REFLECTION EXERCISE

What Makes Data Easy to Understand?

Think about a dataset, package, or repo you found particularly easy (or hard) to use:

GOOD EXPERIENCES

- What information was available?
- Was there a clear, easy-to-find README?

POOR EXPERIENCES

- What vital details were missing?
- How much time did you waste guessing?

EXERCISE FINDINGS

Key Pillars of Usable Data

Explicit Variable Definitions

No ambiguity over column headers, metric scales, and units of measurement.

Collection Context

Information describing experimental conditions, dates, sensors, and sampling sites.

Clear Points of Contact

Named primary investigators or data stewards to resolve queries directly.

Standard Top-Level Placement

Information located immediately where a newcomer expects to look (e.g. root README).

CORE PHILOSOPHY

Documentation Is For Humans

Beyond Technical Infrastructure

People often focus on storage drives, software packages, or file formats.

Yet the single greatest barrier to reusing data is **human comprehension**. Good documentation reduces assumptions.

The Most Important Collaborator

Your most critical collaborator is FUTURE YOU.

Document choices that seem completely obvious today, before memory fades months from now.

DEFINITIONS & PRIVACY

Understanding Metadata

DEFINITION

Data About Data

Transforms meaningless rows of numbers into verifiable scientific findings.

Includes who collected it, dates, instruments, units, licenses, and external sources.



Caution: Hidden Auto-Metadata

Word and Excel automatically embed author names, corporate templates, editing history, and revision numbers.

Always inspect and purge private metadata before sharing files externally.

CORE ELEMENTS

What Should We Document?

Description

Clear project goals and scope.

Provenance

Origin, collections & edits.

Analysis

Software versions & settings.

Contacts

Primary investigators & leads.

README

Top-level guide & structure.

SUMMARISING CONTEXT

Project & Data Descriptions

Project Description

Summarises high-level intent in seconds:

> *"This project investigates disease incidence in marine mammals along the Scottish west coast between 2020 and 2025."*

Far clearer than trying to encode scope into a massive, fragile folder name.

FOLDER PURPOSE DICTIONARY

```
raw_data/
  Original incidence records from collaborators.

processed_data/
  Cleaned, tabular datasets used for modeling.

analysis/
  R scripts and generated figures for publication.
```

PROVENANCE EXERCISE

Challenge: Documenting Data Provenance

For one of your own research projects, what must you document to track provenance?

- Details needed to publish in a journal
- Context required to reproduce the study later

- What if an error is discovered in raw data?
- Can you trace which version corrected it?

PROVENANCE SOLUTION

Provenance Information to Record

Origin & Collection

- **Field/Lab:** When, where, instruments, lab protocols.
- **Collaborator:** Who provided it, and on what date?
- **Repository:** Exact URL, download date, DOI.

Transformation History

- **Software Generation:** Exact version numbers used.
- **Corrections:** What errors were fixed, when, and by whom?
- **Lineage:** Which raw file created which processed table?

EXERCISE & SOLUTION

Documenting Data Analysis & Access Conditions

EXERCISE

Consider what information must be captured about your analytical workflows so others can reproduce your findings.

SOLUTION: VITAL ELEMENTS

- **Software & Packages:** Exact compiler/interpreter versions.
- **Processing Steps:** Settings, filters, and seeds.
- **Assumptions & Limits:** Explicit edge cases.
- **Access Conditions:** Publishing or licensing restrictions.

STANDARD PRACTICE

The Role of the README File

ROOT PLACEMENT

```
project/
├── README.txt    ⟵ First entry point
├── data/
│   ├── raw/
│   └── processed/
├── analysis/
└── results/
```

The Front Door of Your Project

A simple plain-text file placed at the top level explaining what the project is and how to navigate it.

Golden Rule: A simple README with a few useful paragraphs is infinitely better than no README at all!

TEMPLATE DESIGN

Example README Template

```
Project Name: Marine Mammal Incidence Study
Description:  Health records investigation (2020-2025).

Project Structure:
  data/raw       - Original surveyor spreadsheets
  data/processed - Cleaned tabular data
  analysis/      - R scripts for spatial modeling
  results/       - Publication figures & reports

Software: Python 3.13 / pyproject.toml
Contact:  bob.badgerton@example.ac.uk
Updated:  2026-07-01
```

Core Fields Included:

- Title & short synopsis
- Folder directory map
- Dependencies & versions
- Lead contact & last updated date

HANDS-ON EXERCISE

Challenge: Creating a README File

Follow the instructor walkthrough in File Explorer / Finder:

1. Navigate to the top-level `legacy_dataset/` folder.
2. Right-click in an empty area > select **New > Text Document**.
3. Rename the file to `README.txt`.
4. Open in Notepad / TextEdit.
5. Paste core headings: Project Name, Description, Structure, Contact, Updated.

DISCUSSION 1

Documentation Beyond the Root Level

SUBFOLDER README: results/README.txt

```
Generated: 2026-07-01
Software:  R version 4.6
Input:     processed_data_v03.csv
Params:    Significance threshold = 0.05
```

Subfolder Needs

A `results/` or `raw_data/` folder often has distinct execution parameters and source licenses.

👉 **In your own research, what subfolders require dedicated READMEs?**

SPECIALISED FORMATS

Other Useful Documentation Tools

Data Dictionaries
Tabular definitions of variables, allowable ranges, and units.

Standard Operating Procedures (SOPs)
Step-by-step physical or lab processing protocols.

Project Logs
Records of major decisions, abandoned avenues, and rationales.

Parameter Files
Machine-readable configs ensuring execution repeatability.

PRACTICAL EXERCISE

Challenge: Authoring the Inherited README

Fill out the fields in `README.txt` for the inherited project:

- **Project Name & Description**
- **Folder Structure map**
- **Data sources & provenance**
- **Primary contact details**
- **Last updated date**
- Make reasonable, realistic assumptions!

SOLUTION & INVARIANTS

A Complete Project Overview

The Test of a Good README

An external researcher can understand project intent, locate scripts, and know who to contact within **60 seconds** without opening other files.

File-Level Headers

Store critical metadata directly in CSV/data headers.

> **Key Advantage:** In-file metadata cannot be accidentally lost if a file is renamed or moved.

DISCUSSION 2

The Departure Test

Imagine you are leaving your research project tomorrow:

👉 What information would your replacement need to continue seamlessly?

Does your project's current documentation actually contain that information today?

SUMMARY

Episode 4 Key Points

Human Understanding: Documentation makes data understandable, reproducible, and reusable.

Future You: The primary beneficiary of good documentation is your future self.

READMEs: Top-level and subfolder README files act as standard front doors.

In-File Metadata: Headers ensure metadata remains permanently attached to data.

The Invariant: Store documentation as close as possible to the data it describes.

LOOKING AHEAD

Next: Episode 5

Our project now has clean folders, clear filenames, and comprehensive documentation. Next, we address where to store our research data safely and securely.

Choosing Storage Locations